EV Battery Lifecycle Report

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Battery details

Battery type	LFP
Weight (kg)	320.0
Capacity (kWh)	45.0
Cycle count	1,420
State of Health	30.0 %
Remaining Useful Life	0 cycles

Recommendation

Action: RECYCLE **Confidence:** 90%

Reasoning:

- SoH 30.0% below 40%
- RUL 0 cycles critically low

Recoverable materials

Material	Content (kg)	Recovered (kg)	Rate
Li	11.20	10.08	90%
Graphite	64.00	57.60	90%
Cu	25.60	25.09	98%
Al	38.40	37.25	97%

Hazardous: Electrolyte: 32.00 kg, Separator: 6.40 kg

Residual (slag / losses): 121.18 kg

Recommended recycling centres

Centre	Distance (km)	Score
Chennai LFP Recovery	0.0	0.984
Pune Lohum Plant	914.4	0.727
Delhi NCR Battery Recyclers	1,767.7	0.486

Top pick: Chennai LFP Recovery (0.0 km, score 0.984)

Environmental impact

CO2 avoided by recycling this battery: 214 kg

Equivalent to ~856 km of average car driving.

Material	Recovered (kg)	CO2 factor (kg/kg)	CO2 avoided (kg)
Lithium	10.08	5.0	50.40
Graphite	57.60	1.0	57.60
Copper	25.09	2.0	50.18
Aluminium	37.25	1.5	55.88

Emission factors: Argonne GREET model (2023).

Method: RUL estimated via a physics-informed degradation curve (SoH -> remaining cycles). Decision rules based on SoH and RUL bands: CONTINUE_EV (SoH >= 80% and RUL >= 200), SECOND_LIFE (60-80% SoH, RUL >= 50), REPAIR (40-60% SoH), RECYCLE (< 40% SoH or RUL < 50). Materials modelled from chemistry and total battery mass; recovery rates reflect industry-standard pyro + hydro processes.